Diagnostic features:

- Conjoint, Collateral and closed vascular bundle.
- Mesophyll is not differentiated into Palisade and Spongy parenchyma. (Isobilateral leaf)
- Number of Stomata are more or less equal on both epidermis, Stomata surrounded by dumb-bell shaped guard cells.

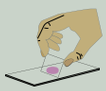
Exercise: 6

Plasmolysis and Deplasmolysis

Aim: Study of plasmolysis in epidermal peel of leaf.

Principal: Living cells are generally turgid due to the presence of water. When cells are immersed in hypertonic solution, shrinkage of protoplasm takes place with visible separation of plasma membrane from the cell walls. This is called plasmolysis and occurs due to exosmosis, a phenomenon in which water from the cells moves into the surrounding medium which is hypertonic, that is more concentrated than the cell sap.

Requirements: Leaves of *Tradescantia*, 70% sugar solution, slide, cover slip, needle, petri dish / watch glass, microscope.



Peel off a small segment from lower epidermal surface of the *Tradescantia* leaf. This can be done by tearing the leaf obliquely with a single jerk or scraping it with blade. Dip it in 70% of sugar solution for 5 minutes. Later mount the peel on a slide to observe plasmolysis.

Again dip the same peel in water for 5 minutes. Later mount it and observe it under the microscope for deplasmolysis.

Diagnostic features: Plasmolysis

- Cell membrane is pulled away from the cell wall.
- Cells becomes flaccid due to loss of water by exosmosis, when a plant cell is kept in a hypertonic solution.

Diagnostic features: Deplasmolysis

- It is reverse of plasmolysis.
- It is swelling of shrinked protoplasm to regain its original unplasmolysed shape when cell is placed in hypotonic solution. It is a type of endosmosis.

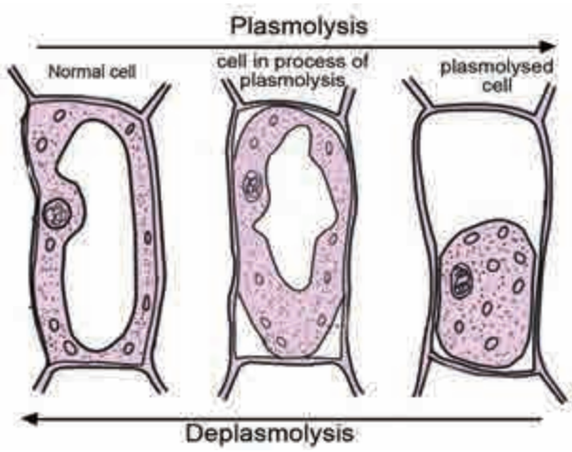


Figure 6: Different Stages of Plasmolysis and Deplasmolysis in a plant cell

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